

Time To Stretch

Context-Aware Stretching Prompts with Movement Verification

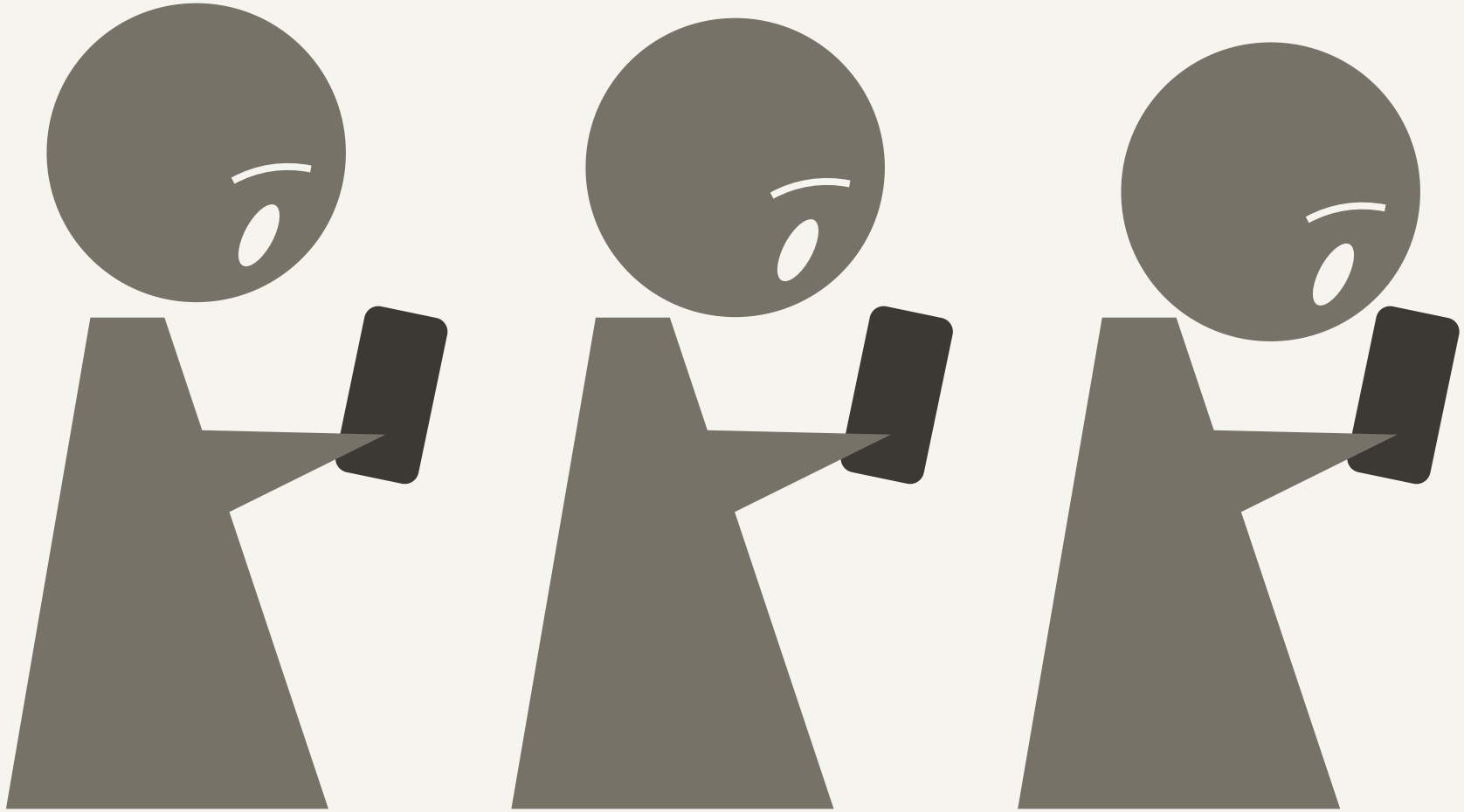


CSED404 Mobile & Ubiquitous Computing

Park Hanbi & Cho Minseo

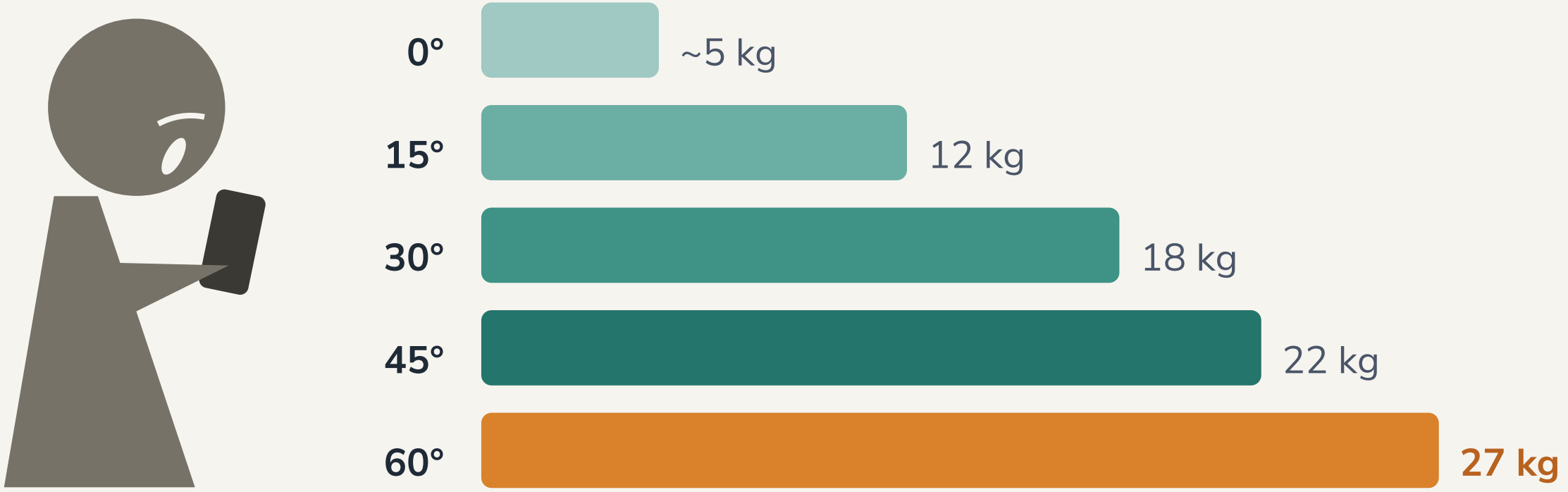


⚠️ **Holding one posture makes us stiff** ⚠️



PROBLEM

Looking down makes your head heavier



[1] Hansraj, K. K. (2014). Surgical Technology International, 25, 277–279.

PROBLEM

Stiff posture adds up to musculoskeletal pain

1.71B

people worldwide live with a musculoskeletal condition

#1

contributor to disability worldwide

160

countries where low back pain is the single leading cause of disability

[3] WHO, Musculoskeletal health fact sheet (GBD 2019 data)

PROBLEM

Stretching works, if it’s regular



Stretch
once in a while

VS

Stretch
short and often



PROBLEM

Stretching works, if it’s regular



Stretch
once in a while

vs

Stretch
short and often



- Less neck and shoulder pain
- Active breaks beat passive rest
- **Consistency drives the effect**

[2] Tunwattanapong et al., Clinical Rehabilitation (2016); Vitoulas et al., Healthcare (2022)

We know it works, but we don’t do it

Unaware

We don’t notice our body stiffening until it already hurts.

Fixed timers

Hourly alarms ignore our actual state and ring mid-class.

No follow-through

Nothing checks whether we stretched, so it never becomes regular.

OUR PROBLEM

How can a phone detect when the body is stiffening and get people to actually stretch, regularly?

EXISTING SOLUTIONS

Stretchly

Scheduled breaks for desktop work

What it offers

- Mini and long breaks
- Activity suggestions
- Idle-time and Do Not Disturb support

Gap for our project

- No documented exercise selection by strain source
- No documented movement verification or outcome-based exercise ranking

[4] hovancik.net/stretchly, github.com/hovancik/stretchly

EXISTING SOLUTIONS

SitApp

Webcam posture feedback with movement breaks

What it offers

- On-device posture monitoring
- Real-time posture nudges
- Seven guided movements in rotation

Gap for our project

- Movement selection follows a preset rotation
- Camera monitoring pauses during breaks
- Users report completion with Done

[5] sitapp.app,

Time To Stretch

CONTEXT-AWARE AND ADAPTIVE STRETCHING RECOMMENDATION SYSTEM

1 Sensing

Phone context

→

2 Estimation

Two debt scores

→

3 Intervention

Timing + exercise

↓

4 Verification

Completion evidence

←

5 User feedback

Difficulty + helpfulness

←

6 Adaptive recommendation

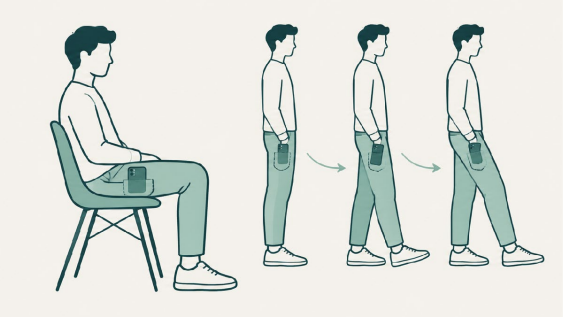
Update exercise priority

- **Estimate accumulated physical strain**
- **Chooses timing and provide personalized exercises**
- **Update future choices by completion evidence and feedback**

METHODOLOGY

Sensing

Target	Signals	Interpretation
Inactivity	Activity Recognition + steps	Elapsed time since detected movement
Phone-use exposure	Screen use + gravity / accelerometer	Duration below a calibrated tilt threshold



Phone carried in pocket

Orientation is a calibrated clue.



Phone held during screen use

Device tilt is a proxy, not a neck-angle measure.

METHODOLOGY

Estimation

Score	Increases when	Decreases when
Neck-related Debt D_n	Low-tilt phone-use exposure continues	Exposure stops or decreases
Sedentary Debt D_s	Inactivity continues	Walking or sufficient activity is detected

$$D = w_n \times D_n + w_s \times D_s$$

METHODOLOGY

Intervention

1 Trigger when D exceeds a calibrated threshold

Defer during detected movement, supported call states, or user-defined class hours. Apply a cooldown between alerts.

Dominant score	Candidate exercises
Higher Neck-related Debt	Neck or shoulder stretches
Higher Sedentary Debt	Walking, back or lower-body movements

Rank suitable candidates using previous completion and user feedback

METHODOLOGY

Verification

Mode	Evidence
Camera enabled	CameraX frames + ML Kit landmarks Custom movement / hold-time rules
Camera-free	Step change + motion App records session duration
Manual confirmation	User reports completion

User Feedback

Question	User input
Exercise difficulty	Easy / Manageable / Difficult
Perceived helpfulness	Helpful / Neutral / Not helpful

Collect information for adaptive recommendation

METHODOLOGY

Adaptive Recommendation

Information	Collection	Use
Completion + evidence type	Verification	Track execution with evidence quality
Difficulty	User feedback	Favor manageable alternatives
Helpfulness	User feedback	Favor positively rated exercises
Session duration	Verification	Analyze session patterns

Case A: stretch A completed

Easy + helpful
Increase A’s priority for similar exposure

Case B: stretch A incomplete

Difficult
Favor an easier stretch B next time

ADVANTAGES

What Makes Us Different?

Cause-Aware Exercise Selection

Preserve neck-related and sedentary exposure separately.
Match the exercise category to the dominant score.

Expected benefit: exercises that better match the current exposure.

Verified & Adaptive Intervention

Combine completion evidence with difficulty and helpfulness ratings.
Update the next exercise choice from the user’s experience.

Expected benefit: more manageable choices and support for participation.

EFFECT

What changes right away

Less frozen time

Shorter spells of static, bent-neck posture

Less stiffness

Targeted stretches for the neck, shoulders and back

Lasting habit

Verified stretching keeps it regular, so it actually works

EFFECT

Beyond computing



Healthcare & rehabilitation

Therapists remotely track daily posture and whether prescribed stretches were done.



Occupational health

Prevents office workers' musculoskeletal pain via wellness programs, cutting lost productivity.



Education

Builds healthy habits for students who study for long hours.

PROJECT ROADMAP

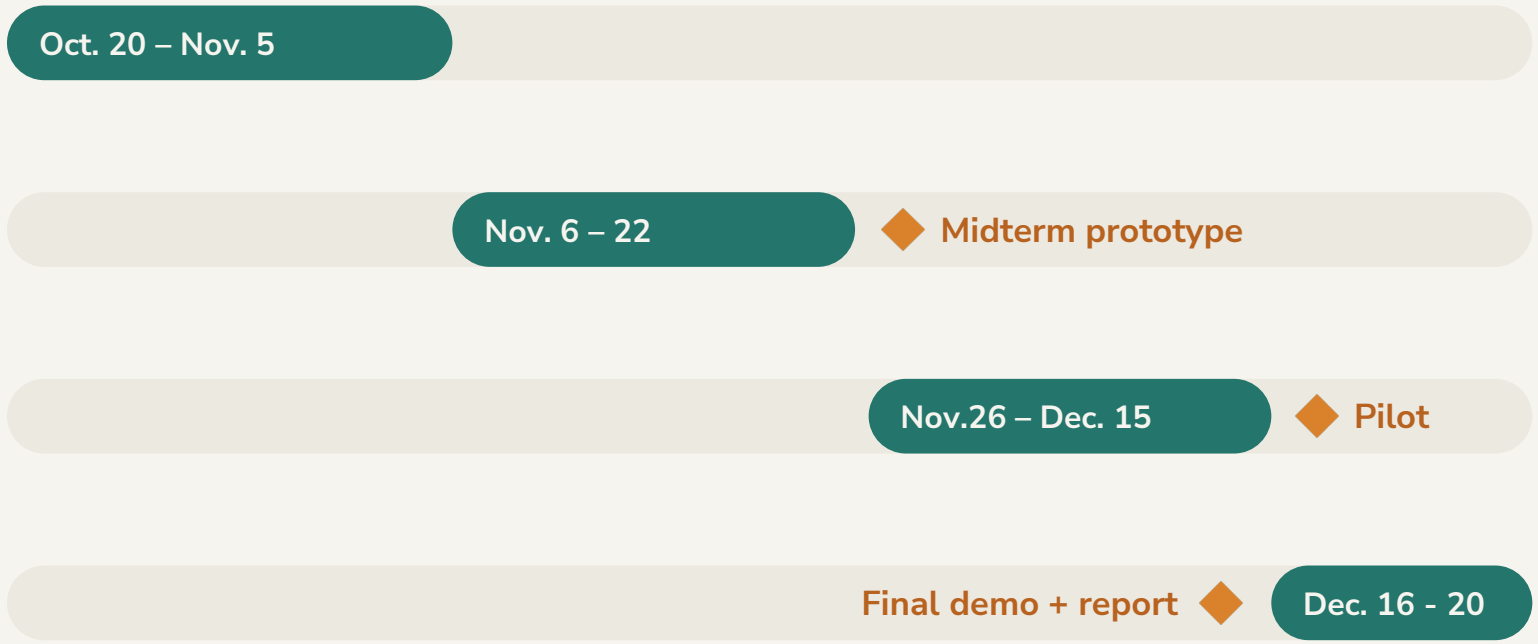
Project Roadmap

Sensor logging
Check phone placement,
missing data and battery cost

Debt scores + alerts
Basic camera / walking verification,
tested against labels

Adaptive ranking + app
Pilot vs. fixed timers:
completion, burden, relevance

Analysis + refinement
Analyze pilot results, refine the prototype



REFERENCES

Research

[1] Hansraj, K. K. (2014). Assessment of stresses in the cervical spine caused by posture and position of the head. *Surgical Technology International, 25*, 277–279.

[2] Tunwattanapong, P., Kongkasuwan, R., & Kuptniratsaikul, V. (2016). The effectiveness of a neck and shoulder stretching exercise program among office workers with neck pain: A randomized controlled trial. *Clinical Rehabilitation, 30*(1), 64–72. https://doi.org/10.1177/0269215515575747

Data

[3] World Health Organization. (2022). *Musculoskeletal health* [Fact sheet]. https://www.who.int/news-room/fact-sheets/detail/musculoskeletal-conditions

Existing Solutions

[4] Hovancik, J. (n.d.). *Stretchly: The break time reminder app*. Retrieved October 2026, from https://hovancik.net/stretchly/

[5] Aston Smith Ltd. (n.d.). *SitApp: Posture Coach* [Mobile app]. App Store. Retrieved October 2026, from https://apps.apple.com/us/app/sitapp-posture-reminder/id6768987893